

RED

⚠ **Security Warning:** Work in an isolated environment (virtual machine or container). Never open unknown files on your primary machine.

Challenge Description

RED, RED, RED, RED

Hints

- The picture seems pure, but is it though?
 - Red?Ged?Bed?Aed?
 - Check whatever Facebook is called now.
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Tools Used

- **file:** Identify image file type and properties
 - **strings:** Extract readable text from binary files
 - **exiftool:** Inspect image metadata and embedded text
 - **zsteg:** Detect steganography in PNG/BMP images
 - **base64:** Decode Base64-encoded data
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Complete Solution

Step 1: Download and Prepare

Download the file named `red.png` from the challenge page.



Step 2: Basic File Inspection

First, verify the file type and general information:

```
file red.png
```

Expected output:

```
red.png: PNG image data, 128 x 128, 8-bit/color RGBA, non-interlaced
```

Step 3: Search for Visible Text

The flag may be present as plain text. Try extracting all character strings:

```
strings red.png | grep -i "picoCTF"
```

⚠ *If no results appear, proceed to the next step (decoding).*

Step 4: Inspect Metadata

Use `exiftool` to examine the file metadata:

```
exiftool red.png
```

Complete output:

```
ExifTool Version Number      : 12.76
File Name                    : red.png
Directory                   : .
File Size                   : 796 bytes
File Modification Date/Time  : 2026:05:08 09:45:07+01:00
File Access Date/Time       : 2026:05:08 09:45:07+01:00
File Inode Change Date/Time  : 2026:05:08 09:45:51+01:00
File Permissions             : -rwxrwxrwx
File Type                   : PNG
```

```
File Type Extension      : png
MIME Type                : image/png
Image Width              : 128
Image Height             : 128
Bit Depth                : 8
Color Type               : RGB with Alpha
Compression              : Deflate/Inflate
Filter                   : Adaptive
Interlace                : Noninterlaced
Poem                    : Crimson heart, vibrant and bold,.Hearts flutter at
your sight..Evenings glow softly red,.Cherries burst with sweet life..Kisses linger
with your warmth..Love deep as merlot..Scarlet leaves falling softly,.Bold in every
stroke.
Image Size               : 128x128
Megapixels               : 0.016
```

The metadata itself is not suspicious, but the `Poem` field contains a hidden clue. Analyze the PNG data again with `strings` to view the full embedded text:

```
strings red.png
```

Output:

```
IHDR
tEXtPoem
Crimson heart, vibrant and bold,
Hearts flutter at your sight.
Evenings glow softly red,
Cherries burst with sweet life.
Kisses linger with your warmth.
Love deep as merlot.
Scarlet leaves falling softly,
Bold in every stroke.x
IDATx
IEND
```

The first letters of each line in the poem spell `CHECK LSB`, so the next step is to analyze the least significant bits.

Step 5: Analyze Steganography with zsteg

Use `zsteg` to inspect the PNG for hidden data in the LSB channels:

```
zsteg red.png
```

Output:

```
meta Poem          .. text: "Crimson heart, vibrant and bold,\nHearts flutter at your sight.\nEvenings glow softly red,\nCherries burst with sweet life.\nKisses linger with your warmth.\nLove deep as merlot.\nScarlet leaves falling softly,\nBold in every stroke."
```

```
chunk:0:IHDR       .. file: Adobe Photoshop Color swatch, version 0, 128 colors; 1st RGB space (0), w 0x80, x 0x806, y 0, z 0; 2nd HSB space (1), w 0x100, x 0, y 0xff01, z 0xff
```

```
b1,rgba,lsb,xy     .. text: "cGljb0NURntyM2RfMXNfdGgzX3VsdDFtNHQzX2N1cjNfZjByXzU0ZG4zNTVffQ==cGljb0NURntyM2RfMXNfdGgzX3VsdDFtNHQzX2N1cjNfZjByXzU0ZG4zNTVffQ==cGljb0NURntyM2RfMXNfdGgzX3VsdDFtNHQzX2N1cjNfZjByXzU0ZG4zNTVffQ==cGljb0NURntyM2RfMXNfdGgzX3VsdDFtNHQzX2N1cjNfZjByXzU0ZG4zNTVffQ=="
```

```
b1,rgba,msb,xy     .. file: OpenPGP Public Key
```

```
b2,g,lsb,xy        .. text: "ET@UETPETUUT@TUUTD@PDUDDPE"
```

```
b2,rgb,lsb,xy      .. file: OpenPGP Secret Key
```

```
b2,bgr,msb,xy      .. file: OpenPGP Public Key
```

```
b2,rgba,lsb,xy     .. file: OpenPGP Secret Key
```

```
b2,rgba,msb,xy     .. text: "CIkiiiII"
```

```
b2,abgr,lsb,xy     .. file: OpenPGP Secret Key
```

```
b2,abgr,msb,xy     .. text: "iiaakikk"
```

```
b3,rgba,msb,xy     .. text: "#wb#wp#7p"
```

```
b3,abgr,msb,xy     .. text: "7r'wb#7p"
```

```
b4,b,lsb,xy        .. file: 0421 Alliant compact executable not stripped
```

The `b1,rgba,lsb,xy` channel contains repeated Base64-encoded text. Decode one instance to reveal the hidden message.

Step 6: Decode the Base64 Payload

```
echo "cGljb0NURntyM2RfMXNfdGgzX3VsdDFtNHQzX2N1cjNfZjByXzU0ZG4zNTVffQ==" | base64 -d
```

✅ Final Result

```
picoCTF{<REDACTED>}
```

Key Concepts

- **Steganography:** Data hidden in image pixels, often using the least significant bits (LSB).
 - **Metadata analysis:** Hidden fields like PNG tEXt chunks can reveal clues.
 - **Base64 encoding:** A common method to hide payloads in readable text.
 - **Tool chaining:** Combine `strings` , `exiftool` , and `zsteg` to uncover hidden content.
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Additional Resources

You can also use online tools to analyze steganography in images:

- [AperiSolve](#) – Online steganography analysis tool